

One talk a month - from symplectic to toric manifolds

Talks by [Rupadarshi Ray](#) in the discord server "IISERs Math Forum"

1. 26 June, '25 - Introduction to differential forms, Hamiltonian flows, cohomology
2. 26 July, '25 - Irrational torus translations
3. More on symplectic manifolds, symplectic actions
4. Moment maps and Hamiltonian actions by Abelian Lie groups
5. More moment maps
6. Luoville-Arnold theorem
7. Toric manifolds
8. ... Toric varieties!

derivative and integral of differential forms

$$\int f \, dx \, dy = \int f \, \boxed{r} \, dr \, d\theta$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx = -r \sin \theta \, d\theta + \cos \theta \, dr$$

$$dy = r \cos \theta \, d\theta + \sin \theta \, dr$$

$$dx \wedge dy = r \cos^2 \theta \, dr \wedge d\theta + r \sin^2 \theta \, dr \wedge d\theta$$

$$= \boxed{r} \, dr \, d\theta$$

motivation

We look at the (familiar) structure of

$$\mathcal{C}^\infty(U) \xrightarrow{\text{grad}} \text{Vec}^\infty(U) \xrightarrow{\text{curl}} \text{Vec}^\infty(U) \xrightarrow{\text{div}} \mathcal{C}^\infty(U)$$

$$\text{curl} \circ \text{grad} = 0$$

$$\text{div} \circ \text{curl} = 0$$

where $U \subseteq \mathbb{R}^3$.

Writing this in a different notation: for $\mathcal{C}^\infty(U) \xrightarrow{\text{grad}} \text{Vec}^\infty(U) \xrightarrow{\text{curl}} \text{Vec}^\infty(U)$

$$f \xrightarrow{\text{d}} \text{d}f = \underbrace{\sum_i \frac{\partial f}{\partial x_i} \text{d}x_i}_{\text{this looks like grad}}$$

$$\omega = \sum_i \omega_i \text{d}x_i \xrightarrow{\text{d}} \text{d}\omega = \underbrace{\sum_i \text{d}\omega_i \wedge \text{d}x_i}_{\text{this looks like curl}}$$

and then continuing for $\text{Vec}^\infty(U) \xrightarrow{\text{div}} \mathcal{C}^\infty(U)$

$$\text{d}\omega = \underbrace{\sum_i \text{d}\omega_i \wedge \text{d}x_i}_{\text{this looks like curl}}$$

$$\alpha = \sum_{i>j} \alpha_{ij} \text{d}x_i \wedge \text{d}x_j \xrightarrow{\text{d}} \text{d}\alpha = \underbrace{\sum_{i>j} \text{d}\alpha_{ij} \wedge \text{d}x_i \wedge \text{d}x_j}_{\text{this looks like div (check!)}}$$

$$\beta = \beta_{123} \text{d}x_1 \wedge \text{d}x_2 \wedge \text{d}x_3 \xrightarrow{\text{d}} 0$$

where we understand that the "wedge" $\wedge$ works like the cross product

$$\text{d}x_i \wedge \text{d}x_j = -\text{d}x_j \wedge \text{d}x_i$$

It is easily seen

$$\text{d}(\text{d}(-)) = 0$$

doesn't matter what's inside the $(-)$, a real function f or the objects ω , α or β (called 0,1,2,3-forms respectively) when their components are differentiable functions.

We see this "exterior derivative"

$$d$$

thus "unifies" grad, curl and div and generalizes because it may be defined as

$$d\omega = d\left(\sum \omega_I dx_I\right) := d\omega_I \wedge dx_I$$

in any dimension.

But what are these

$$dx_i$$

mean?

derivative gives you chain complex

$$\omega = \sum w_i dx_i \leftarrow 1\text{-form}$$

$$d\omega = \sum dw_i \wedge dx_i$$

$$= \sum_{i,j} \frac{\partial w_i}{\partial x_j} dx_j \wedge dx_i$$

$$= \sum_{(i,j)} \underbrace{\left(\frac{\partial w_i}{\partial x_j} - \frac{\partial w_j}{\partial x_i} \right)} dx_j \wedge dx_i$$

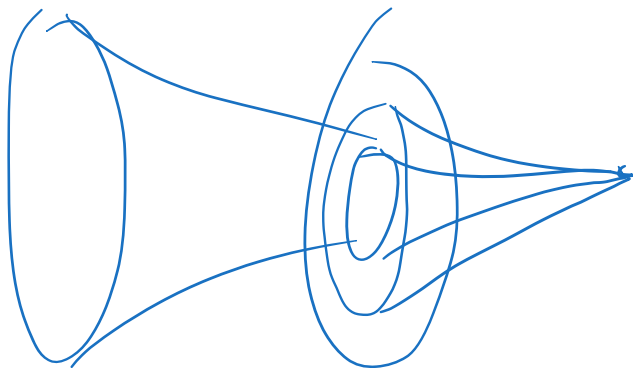
$$0 \rightarrow \underbrace{\text{loc } \omega^k}_\omega \xrightarrow{d} \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \dots \xrightarrow{d} \Omega^n \xrightarrow{d} 0$$

$$\mathbb{R} \xrightarrow{d} \sum \frac{\partial f}{\partial x_i} dx_i \rightarrow 0$$

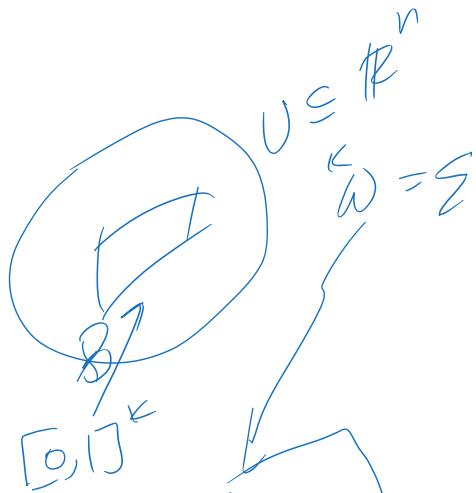
$$\sum_n w_i dx_i \xrightarrow{\quad} \sum \underbrace{\alpha_{ij} dx_i \wedge dx_j}_{n_{C_2}} \rightarrow 0$$

$$\underbrace{\wedge dx_1 \wedge \dots \wedge dx_n}_{\quad}$$

$$d \circ d = 0$$

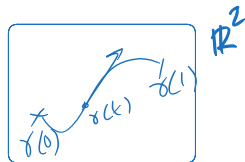


integration gives you pairing



$$\int_{[0,1]^k} y^*(\omega)$$

$$\underbrace{dx}_k$$



$$\omega = \omega_1 dx_1 + \omega_2 dx_2$$

$$= \omega_1 \dot{\gamma}_1 + \omega_2 \dot{\gamma}_2$$

$$\gamma: [0,1] \rightarrow \mathbb{R}^2$$

$$\int_{\gamma} \omega := \int_{[0,1]} \omega_{\gamma(t)}(\dot{\gamma}(t))$$

$$\dot{\gamma} = \begin{bmatrix} \dot{\gamma}_1 \\ \dot{\gamma}_2 \end{bmatrix}$$

$$dx = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$dx(\dot{\gamma}) = \dot{\gamma}_1$$

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\int_{\gamma} df = f(\gamma(1)) - f(\gamma(0))$$

$$= \int_{\gamma} f_{\partial \gamma}$$

$$\int_{\gamma_B} d(\omega) = \int_{\partial B} \omega$$

on $\mathbb{R}^n$

the cohomology of differential forms is topological

☰ (de Rham isomorphism)

$$H^k(\Omega^\bullet(M), d) = H^k(M, \mathbb{R})$$

flows of vector fields motivating Hamiltonian flows

on the plane

We want a vector field that *preserves* a given smooth

$$H : \mathbb{R}^2 \rightarrow \mathbb{R}$$

Then

$$\text{grad}(H)$$

does the opposite: its flow goes out of $H^{-1}(c)$ *curves*.

So we should work with the 90 degrees rotation of the gradient, so

$$J\text{grad}(H)$$

where

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is the clockwise 90 degrees rotation matrix.

The flow of this vector field ϕ^t such that

$$\frac{d}{dt}\phi^t(p) = J\text{grad}(H)_{\phi^t(p)}$$

which quickly implies

$$\frac{d}{dt}H(\phi^t(p)) = dH\left(\frac{d}{dt}\phi^t(p)\right) = \langle \text{grad}(H), J\text{grad}(H) \rangle = 0$$

by chain rule!

Hamiltonian vector fields on plane preserve area

$$X_H = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

$$\operatorname{div}(X_H) = -\frac{\partial}{\partial x} \frac{\partial H}{\partial y} + \frac{\partial}{\partial y} \frac{\partial H}{\partial x} = 0$$

And divergence free vector fields preserve area (volume in $\mathbb{R}^n$)

are all area preserving flows Hamiltonian vector fields?

Well,

- $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is *Hamiltonian* for some smooth H

- $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

- $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : dH = X_2 dx - X_1 dy$$

- So we may use facts about 1-forms

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \leftrightarrow \alpha = X_2 dx - X_1 dy$$

- We know **on a simply connected domain** α is closed ($d\alpha = 0$) $\iff$ α is exact
 $\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \alpha = dH.$

- Now

$$d\alpha = \underbrace{\left(-\frac{\partial X_1}{\partial x} - \frac{\partial X_2}{\partial y} \right)}_{-\text{div}(X)} dx \wedge dy$$

- Hence, on a **simply connected domain** flow of X is area preserving $\iff X$ is divergence free $\iff \exists H \in \mathcal{C}^\infty(\mathbb{R}^2)$ such that

$$X = J\text{grad}(H)$$

-  Notice a minus sign pops out in $d\alpha$ above.

isomorphism of space of symplectic modulo Hamiltonian vector fields with first cohomology

Proposition:

$$\frac{\text{Sym}(M)}{\text{Ham}(M)} \cong H^1(\Omega^\bullet, d) \cong H^1(M, \mathbb{R})$$

summary

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) <i>symplectic manifolds</i>
$J = [i]$	J is the complex structure on the Riemann sphere	$J = \begin{bmatrix} i & & \\ & i & \\ & & \ddots \end{bmatrix}$	We don't need it in general!
$X_H := J \text{grad}(H)$	$X_H := J \text{grad}(H)$	$X_H := J \text{grad}(H)$	$X_H := -\omega^{-1} dH$ <i>notice the "-" sign!</i>
$X_H H = 0$	$X_H H = 0$	$X_H H = 0$	$X_H H = 0$
<i>area preserving</i> $\mathcal{L}_{X_H} dx \wedge dy = 0$	<i>area preserving</i> $\mathcal{L}_{X_H} \lambda_{\text{area}} = 0$	volume preserving, but preserves more...	<i>"k-dimensional area preserving"</i> $\mathcal{L}_{X_H} \omega^k = 0$ for every $1 \leq k \leq n$
Converse? True for $\mathbb{R}^2$ but not true for general <i>open subsets!</i>	Converse? true for S^2 !	Converse for volume preservation is false for $n > 2$, even for linear flows!	Converse is false for $\dim M > 2$! In higher dimensions Hamiltonian flows preserve ω^k for all $1 \leq k \leq n$ so preserving ω becomes a necessary condition. However, it is true that vector fields that preserve the <i>symplectic 2-form</i> are Hamiltonian on a simply connected M .

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) <i>symplectic manifolds</i>
Linear maps that are <i>generated</i> by Hamiltonian vector fields are just the area preserving ones $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$	What could be "linear" flows? well isometries $SO(3, \mathbb{R})$ preserve area, so we have one nice Lie group action on the 2-sphere.		symplectic G-actions, Hamiltonian G-actions
Is every area preserving flow Hamiltonian? Yes But not true for the space $\mathbb{R}^2 \setminus \{0\}$	Is every area preserving flow Hamiltonian? Yes	Is every symplectic flow Hamiltonian? Yes	Is every symplectic flow Hamiltonian? We have the isomorphism $H^1(M, \mathbb{R}) \cong H^1(\Omega^\bullet, d) \cong \frac{\text{Sym}(M, \omega)}{\text{Hamil}(M, \omega)}$

motivating torus translations and diagonal action on $\mathbb{C}^2$

Consider the Lissajous figures:

✓ <https://www.desmos.com/calculator/8xuzb23i42>



$AC \mathbb{R}^2$ is
dense if

$x \in \mathbb{R}^2$



$B(x, \epsilon) \cap A \neq \emptyset$

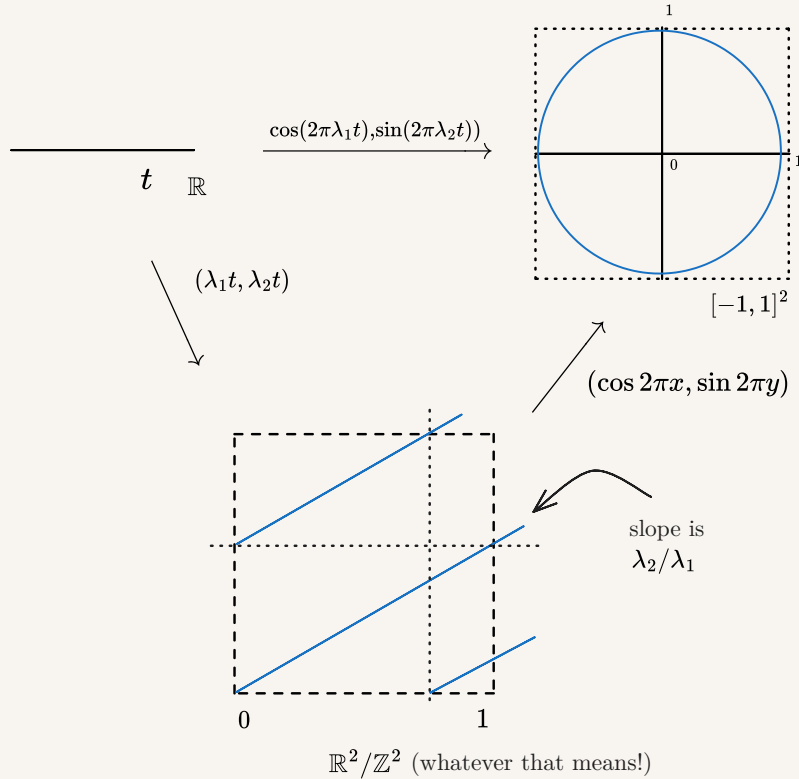
the Lissajous figure way

Proposition: Show that $\frac{\lambda_1}{\lambda_2}$ is irrational $\iff$

$$\{(\cos(2\pi\lambda_1 t), \sin(2\pi\lambda_2 t)) : t \in \mathbb{R}\}$$

is dense in $[-1, 1]^2$.

simplify



Proposition:

$$\Phi := (\cos 2\pi x, \sin 2\pi y) : \mathbb{R}^2 / \mathbb{Z}^2 \rightarrow [-1, 1]^2$$

is surjective continuous.

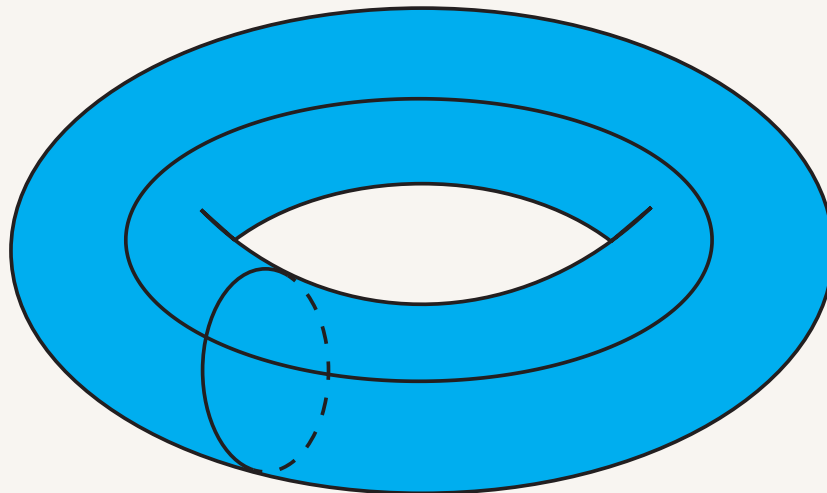
work on the simplification

Proposition:

$$\{(n\lambda_1, n\lambda_2) \bmod \mathbb{Z}^2 : n \in \mathbb{Z}\} \subset \mathbb{R}^2 / \mathbb{Z}^2$$

is dense in $\mathbb{R}^2 / \mathbb{Z}^2$.

Here, $\mathbb{R}^2 / \mathbb{Z}^2$ is the torus!



we need to show that the "lines" on the $\mathbb{R}^2/\mathbb{Z}^2$ is a dense subset iff its slope is irrational

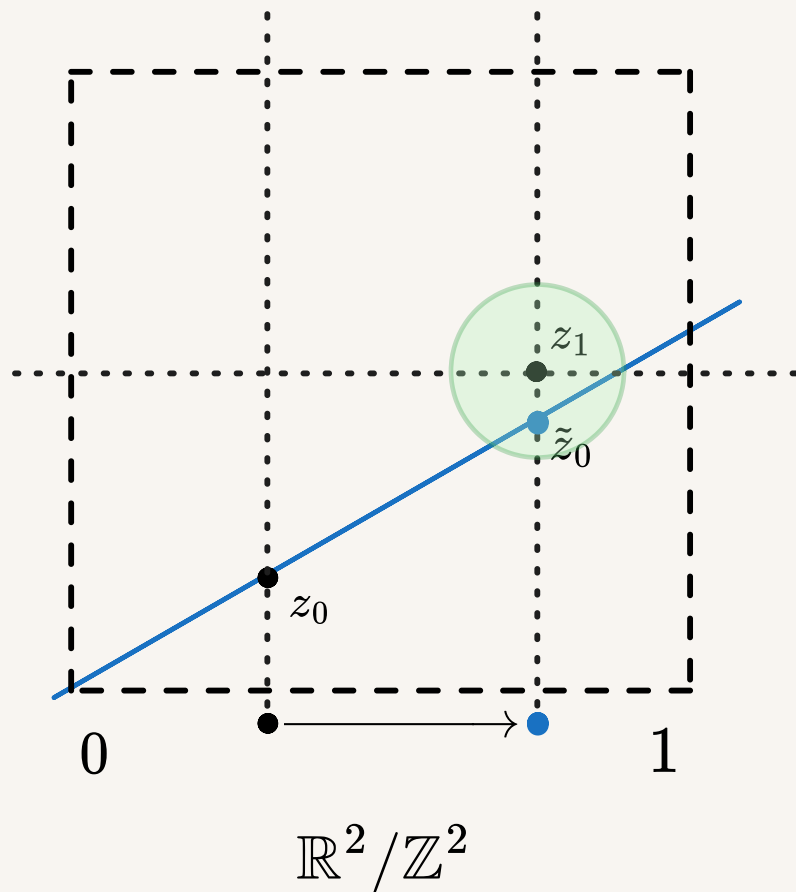
Start anywhere, say z_0 , then the orbit of z_0 should come ϵ -close to any other point z_1 .

The flow looks like

$$(\theta_1, \theta_2) \xrightarrow{t} (\theta_1 + t \bmod 1, \theta_2 + lt \bmod 1)$$

We can always match the first coordinate of $z_0 \equiv (\phi_1, \phi_2)$ by

$$(\theta_1, \theta_2) \xrightarrow{t=\phi_1-\theta_1+n} (\phi_1, \theta_2 + l(\phi_1 - \theta_1 + n) \bmod 1)$$



So for $n \geq 1$ we reach the points (ϕ_1, c_n) where

$$c_n = c_0 + ln \bmod 1$$

We must show there is one point which is ϵ -close to (ϕ_1, ϕ_2)

space.R.sphere.1.End rotation

Left translation $S^1 \curvearrowright S^1$, **AKA** rotations on the circle

 **Definition.** Left translation $S^1 \curvearrowright S^1$, **AKA** rotations on the circle

The left translation of the circle group (by the element $\exp(i2\pi\theta) \in S^1$) S^1

$$\begin{aligned} S^1 &\rightarrow S^1 \\ z &\mapsto \exp(i2\pi\theta)z \end{aligned}$$

for $\theta \in \mathbb{R}$ is also called a **rotation** on the circle.

If $\theta \in \mathbb{Q}$ the transformation is called a **rational rotation**, that is, the action by $\mathbb{Q}/\mathbb{Z} \subset \mathbb{R}/\mathbb{Z} \cong S^1$.

If $\theta \in \mathbb{R} \setminus \mathbb{Q}$, then its a **irrational rotation**.

rational rotations are periodic

Consider the *rational* rotation

$$R^{\frac{p}{q}}(z) := \exp\left(2\pi i \frac{p}{q}\right)z$$

where $p, q \in \mathbb{Z}_{>0}$. After q many iterations we have

$$\left(R^{\frac{p}{q}}\right)^q(z) = \exp(2\pi i p)z = z$$

Thus every point in S^1 is periodic under iterations of $R^{\frac{p}{q}}$.

For any rotation by $\exp(i\theta)$ if

$$\exp(i\theta)^k = 1 \implies k\theta \in 2\pi\mathbb{Z} \implies \frac{\theta}{2\pi} \in \mathbb{Q}$$

irrational rotations are minimal

Consider the rotation

$$R^\theta(z) := \exp(2\pi i \theta)z$$

where $\theta \in \mathbb{R} \setminus \mathbb{Q}$ (irrational).

- If there is any $z \in S^1$ with $k > 1$ such that

$$\begin{aligned} (R^\theta)^k(z) &= z \\ \implies \exp(2\pi i k \theta) z &= z \\ \implies \exp(2\pi i k \theta) &= 1 \\ \implies k\theta &\in \mathbb{Z} \\ \implies \theta &\in \mathbb{Q} \end{aligned}$$

which is a contradiction to $\theta \in \mathbb{R} \setminus \mathbb{Q}$. Therefore, there are no periodic orbits of irrational rotation R^θ .

- Points are not periodic $\iff$ orbit A_θ is infinite as rotations are bijective.

Proposition: Consider the forward orbit of $1 \in S^1$, that is,

$$A_\theta := \{(R^\theta)^k(1) = \exp(2\pi i \theta k) \mid k \in \mathbb{N}\}$$

and any other forward orbit is a translate of A_θ

$$\{(R^\theta)^k(z) = \exp(2\pi i \theta k) z \mid k \in \mathbb{N}\} = A_\theta z$$

The set

$$A_\theta \subset S^1$$

is **dense** (and therefore $A_\theta z$ is also dense in S^1).

Therefore, every orbit of an irrational rotation is dense in the circle making irrational rotations *minimal*.

proof by circle partitions

☀ *Assume* A_θ is not dense in the circle S^1 , so there is an interval $I \subset S^1$ of size $\epsilon > 0$ which does not intersect A_θ .

- *Assume* there are $m > n$ such that

$$d(\exp(2\pi i \theta m), \exp(2\pi i \theta n)) = d < \epsilon$$

- Then rotating by $(R^\theta)^{m-n}$

$$d(\exp(2\pi i \theta (m - n))z, z) = d < \epsilon$$

- So for any $z \in A_\theta$, we may partition $S^1 \setminus I$ into roughly $(1 - \epsilon)/d$ intervals of size d (and one a little smaller) and put some finite iteration of z into I .
- This contradicts I not intersecting A_θ , this there is no such m, n , implying points in A_θ at the least ϵ distance apart.
- But, that would mean there are at most $1/\epsilon$ elements in A_θ which contradicts the orbit A_θ being infinite.

• Points are not periodic $\iff$ orbit A_θ is infinite as rotations are bijective.

Alternatively, we may construct a sub-orbit with consecutive distance $< \epsilon$:

☀ Consider an arbitrary integer $N > 0$ and the N arcs of length $\frac{1}{N}$ of $\mathbb{R}/\mathbb{Z}$ with boundary points

$$0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}$$

- Since the orbit A_θ is infinite there exists integers $n_1 < n_2$ such that

$$|(n_1 - n_2)\theta| \bmod 1 < \frac{1}{N}$$

- Thus for every $\epsilon > 0$ there exists $n_\epsilon \in \mathbb{Z}$ such that

$$n_\epsilon \theta \bmod 1 < \epsilon$$

- Thus the distance between the consecutive entries

$$0, n_\epsilon \theta \bmod 1, 2n_\epsilon \theta \bmod 1, 3n_\epsilon \theta \bmod 1, \dots$$

is $< \epsilon$. Thus it must require finitely many (roughly $\frac{1}{\epsilon}$) entries from the front to partition $\mathbb{R}/\mathbb{Z}$ into size length $< \epsilon$.

- Any $\alpha \in \mathbb{R}/\mathbb{Z}$ must be in one of those partitions, thus there is a k such that

$$|\alpha - kn_\epsilon \theta| < \epsilon$$

- Thus A_θ is dense in $\mathbb{R}/\mathbb{Z}$.

Alternative, even simply we can argue about the complement of the closure of the orbit

- ☀ Let $T(z) = \lambda z$ be an irrational circle rotation. Consider an orbit $X \subset S^1$, and the complement of its closure

$$Y := S^1 \setminus \bar{X}$$

- *Assume* Y is not empty. Because $\bar{X}$ is invariant and closed, Y is invariant and open.
- Any open subset of S^1 is disjoint union of intervals. Consider the collection of disjoint intervals $\{L_1, \dots, L_n\}$ of a fixed length. There must be finitely many of them as total length of S^1 is finite.
- Because rotation is length preserving, $\bigcup_i L_i$ must be an invariant set, and because these sets are disjoint, the rotation must permute these sets.
- So there is a $k > 0$ such that $T^k(L_1) = L_1$
 - that is the rotation T^k on $\mathbb{R}^2$ must preserve the proper arc $L_1 \subset S^1$
 - *OR* the endpoint of L_1 would map to itself, meaning its a fixed point of T^k , but the only rotation with a fixed point is the identity
- Thus $T^k = \text{Id}$ making T periodic, which contradicts it being irrational.
- Hence, Y is empty, making every orbit dense $\iff$ no proper closed non-empty subset.

proof by dense subgroups

- Consider

$$B_\theta := \{n\theta + m : n, m \in \mathbb{Z}\}$$

then

$$B_\theta \xrightarrow{\exp(2\pi i -)} A_\theta$$

- Clearly, $(B_\theta, +)$ is a subgroup of $(\mathbb{R}, +)$.

- If $0 \in B_\theta$ is a *limit point* of B_θ , then there is a $x \in B_\theta$ such that

$$0 < x < \epsilon$$

- Then for any $y \in \mathbb{R}$, there exists $n \in \mathbb{Z}$ such that

$$nx \leq y < (n+1)x$$

which shifts to

$$y - nx < x < \epsilon$$

making nx

ergodicity

spectrum of circle rotations

We have the unitary map $R^{\theta*}$ on $L^2(S^1)$ such that

$$R^{\theta*}(f)(z) := f(R^\theta(x))$$

On the Fourier basis $(e_k)_{k \in \mathbb{Z}}$ this acts by

$$R^{\theta*}(e_k) = \exp(2\pi i \theta)^k e_k$$

so it is diagonal with eigenvalues A_θ .

- If θ is rational, then its spectral measure is

$$\mu = \sum_{x \in A_\theta} \delta_x$$

with multiplicity $M|_{A_\theta} \equiv \infty$.

- If θ is irrational then its spectral measure is

$$\mu = \sum_{x \in A_\theta} \delta_x$$

with multiplicity $M|_{A_\theta} \equiv 1$

☐ **An irrational rotation of the circle has *pure point spectrum of multiplicity 1* supported on $\langle \lambda \rangle < S^1$ for some $\lambda \in S^1$ of infinite order that is either the rotation angle or its inverse. Thus, two irrational rotations by $\lambda_1, \lambda_2 \in S^1$ are isomorphic $\iff \lambda_1 = \lambda_2^\pm$.**

A better way to look at this is

$$A \in M_n(\mathbb{R})$$

$$\exp(tA) : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$\begin{aligned} \exp(t_1 A) \exp(t_2 A)(p) \\ = \exp((t_1 + t_2)A)(p) \end{aligned}$$

$$\mathbb{R} \sim \mathbb{R}^n$$

$$\boxed{\begin{array}{c} \mathbb{R} \sim \Phi^2 \\ (z_1, z_2) \xrightarrow{t} \begin{bmatrix} e^{i\omega_1 t} & \\ & e^{i\omega_2 t} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \Phi^t \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \end{array}}$$

$$\{z_1 \bar{z}_1 + z_2 \bar{z}_2 \mid (z_1, z_2) \in \mathbb{C}^2\} = S^3$$

$$(z_1, z_2) \in S^3 \xrightarrow{t}$$

$$\begin{array}{l|l} H_1(z_1, z_2) = z_1 \bar{z}_1 & H_0(\Phi^t(z_1, z_2)) \\ H_2(z_1, z_2) = z_2 \bar{z}_2 & = H_1(z_1, z_2) \end{array}$$

$$(z_1, z_2) \in H_1^{-1}(c_1) \cap H_2^{-1}(c_2)$$

$$\left\{ z_1 \bar{z}_1 = r_1^2, z_2 \bar{z}_2 = r_2^2 \mid \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{C}^2 \right\}$$

$$r_1, r_2 > 0 \downarrow \parallel$$

$$S^1 \times S^1$$

$$\left(\begin{array}{c} i\theta_1 \\ i\theta_2 \end{array} \right)$$

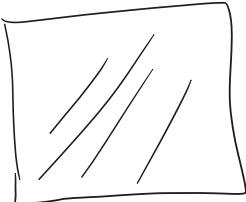
$$(e^{i\theta_1}, e^{i\theta_2}) \rightarrow (\underbrace{\gamma_1 e^{i\theta_1}}_{s'_1}, \underbrace{\gamma_2 e^{i\theta_2}}_{s'_2})$$

$$s'_1 \times s'_2 \longrightarrow T \subset \mathbb{C}^2$$

$$\forall \gamma_1, \gamma_2 > 0$$

$$T_{\gamma_1, \gamma_2} \subset \mathbb{C}^2$$

$$\begin{pmatrix} \gamma_1 e^{i\theta_1} \\ \gamma_2 e^{i\theta_2} \end{pmatrix} \xrightarrow{t} \begin{pmatrix} \gamma_1 e^{i(\theta_1 + t\omega_1)} \\ \gamma_2 e^{i(\theta_2 + t\omega_2)} \end{pmatrix}$$

$$\mathbb{R} \curvearrowright S'_1 \times S'_2$$


$$\mathbb{R} \curvearrowright \mathbb{C}^2$$

$$\sim \cup$$

$$\downarrow \underline{T_{r_1 r_2}}$$

$$H := \frac{1}{2} (\omega_2 \underline{z_1} \bar{z_1} + \omega_1 \underline{z_2} \bar{z_2})$$

$$\left[\begin{array}{c} T^2 \curvearrowright \mathbb{C}^2 \\ (e^{i\theta_1}, e^{i\theta_2}) \mapsto \begin{bmatrix} e^{i\theta_1} & e^{i\theta_2} \end{bmatrix} \\ \mathbb{R}_t \longrightarrow T^2_{(e^{i\theta_1}, e^{i\theta_2})} \end{array} \right]$$

a Hamiltonian
action

space.f.oscillators.harmonic.2

diagonal $T^2 \curvearrowright \mathbb{C}^2$

AKA the dynamics of 2 harmonic oscillators

On the symplectic 4-manifold

$$\left(\mathbb{R}^4, \sum_{i=1}^2 dp_i \wedge dq_i \right) = \left(\mathbb{C}^2, \frac{1}{2i} \sum_{i=1}^2 dz_i \wedge d\bar{z}_i \right)$$

we have the following Hamiltonian vector field and flow

equations	vector field	flow	Hamiltonian
$\begin{aligned}\dot{p}_1 &= -a_1 q_1 \\ \dot{q}_1 &= a_1 p_1 \\ \dot{p}_2 &= -a_2 q_2 \\ \dot{q}_2 &= a_2 p_2\end{aligned}$ or in more revealing, matrix form $\begin{bmatrix} \dot{p}_1 \\ \dot{q}_1 \\ \dot{p}_2 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} & -a_1 & & \\ a_1 & & & \\ & & & -a_2 \\ & & a_2 & \end{bmatrix}$ where the matrix is a block matrix with a scaled $\pi/2$-rotation matrix	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1}$		$\omega_1 \underbrace{\frac{p_1^2 + q_1^2}{2}}_{H_1} + \omega_2 \underbrace{\frac{p_2^2 + q_2^2}{2}}_{H_2}$

equations	vector field	flow	Hamiltonian
$\begin{aligned}\dot{z}_1 &= i\omega_1 z_1 \\ \dot{z}_2 &= i\omega_2 z_2\end{aligned}$ or $\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} i\omega_1 & \\ & i\omega_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$	$\underbrace{\omega_1 i z_1 \frac{\partial}{\partial z_1}}_{X_1} + \underbrace{\omega_2 i z_2 \frac{\partial}{\partial z_2}}_{X_2}$ $= \omega_1 X_1 + \omega_2 X_2$	$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \xrightarrow{t} \begin{bmatrix} \exp(i\omega_1 t) z_1 \\ \exp(i\omega_2 t) z_2 \end{bmatrix}$	$\omega_1 \underbrace{\frac{z_1 \bar{z}_1}{2}}_{H_1} + \omega_2 \underbrace{\frac{z_2 \bar{z}_2}{2}}_{H_2}$

We may rephrase in terms of the diagonal action on $\mathbb{C}^2$ by $\mathbb{T}^2$

action	generating vector field	moment map
diagonal $\mathbb{T}^2$-action on $\mathbb{C}^2$ <i>with both frequencies = 1</i> given by $\mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \xrightarrow{t_1, t_2} \begin{bmatrix} \exp(it_1) z_1 \\ \exp(it_2) z_2 \end{bmatrix}$	the generators of the $\mathbb{T}^2$-action are the vector fields X_1, X_2 with $[X_1, X_2] = 0$	the H_1, H_2 above with $\{H_1, H_2\} = 0$ give with moment map $(H_1, H_2) : \mathbb{C}^2 \rightarrow \mathbb{R}^2$ $(z, z_2) \mapsto \frac{1}{2}(z_1 \bar{z}_1, z_2 \bar{z}_2)$ of the Hamiltonian $\mathbb{T}^2$-action

action	generating vector field	moment map
from the homomorphism $S^1 \rightarrow \mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $t \mapsto l_1 t + l_2 t$ for $l_1, l_2 \in \mathbb{Z}$ we get a S^1-action on $\mathbb{C}^2$ with <i>integer frequencies</i>	the generator is $l_1 X_1 + l_2 X_2$	
$\mathbb{R} \rightarrow \mathbb{T}^2$	$\omega_1 X_1 + \omega_2 X_2$	
$\mathbb{T}^2 \rightarrow \mathbb{T}^2$		

restricting to the invariant 3-sphere

We restrict to the 3-sphere

$$S_h^3 = \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \cong_{\text{Man}} S^3$$

which is invariant under the flow of $\omega_1 X_1 + \omega_2 X_2$ and has orbits of $H_1 = h$. On this sphere, we have the invariant tori

$$T_{h_1, h_2}^2 := \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2h_1, p_2^2 + q_2^2 = 2h_2\}$$

with $\omega_1 h_1 + \omega_2 h_2 = h$.

The projection

$$S^3 \setminus \{(0, 0, 0, 1)\} \rightarrow \mathbb{R}^3$$
$$(x, y, z, w) \mapsto (X, Y, Z) := \frac{1}{1-w}(x, y, z)$$

maps

- the orbit

$$(p_1, q_1, 0, 0) \mapsto (p_1, q_1, 0)$$

where $p_1^2 + q_1^2 = 2h_1$, the image is thus a circle in the x-y plane in $\mathbb{R}^3$

- the orbit

$$(0, 0, p_2, q_2) \mapsto \left(0, 0, \frac{p_2}{1-q_2}\right)$$

whose image is the z-axis

- $T_{h_1, h_2}^2 \mapsto \left\{ (X, Y, Z) \left| X^2 + Y^2 = \frac{2h_1}{(1-w)^2}, Z^2 = \frac{2h_2 - w^2}{(1-w)^2}, w \neq 1 \right. \right\}$

which should be a torus in $\mathbb{R}^3$

-

restricting to the invariant tori: a torus translation

The subset

$$\begin{aligned}T_{a,b}^2 &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2a, p_2^2 + q_2^2 = 2b\} \subseteq (\mathbb{R}^2 \setminus \{0\})^2 \\ S_h^3 &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \subseteq \mathbb{R}^4 \setminus \{0\}\end{aligned}$$

which is preserved by the flow of $\omega_1 X_1 + \omega_2 X_2$.

Thus $(\mathbb{R}^2 \setminus \{0\})^2$ has a torus fibration



coordinates	X_1	$\omega_1 X_1 + \omega_2 X_2$	flow
On $\mathbb{R}^4$ we have (p_1, q_1, p_2, q_2)	$-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}$	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1}$	
On $(\mathbb{R}^2 \setminus \{0\})^2$ we have the coordinates $r_1^2 = p_1^2 + q_1^2$ θ_1 $r_2^2 = p_2^2 + q_2^2$ θ_2	$\frac{\partial}{\partial \theta_1}$	$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$ which is a torus translation $T_{a,b}^2$	

Flow of a general vector field

$$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$$

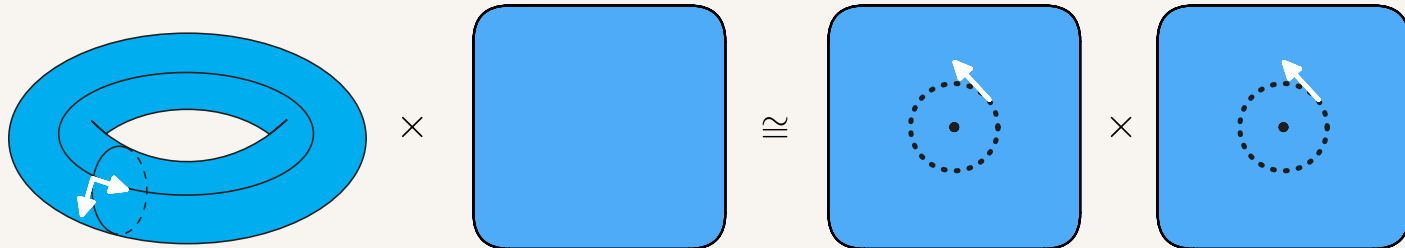
is given by

$$\exp \left(t \left(\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2} \right) \right) \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + t \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

whose flows have properties

if	flow is	which gives a	that is coming from
$\omega_1/\omega_2 \in \mathbb{Q}$	periodic	S^1 -action on T^2	composition $S^1 \hookrightarrow T^2 \curvearrowright T^2$
$\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$	not periodic, minimal (orbits are dense)	$\mathbb{R}$ -action on T^2	composition $\mathbb{R} \hookrightarrow T^2 \curvearrowright T^2$

decomposition into torus translation $\times$ trivial



$$\frac{\partial}{\partial \theta_1}$$

$$0$$

$$\mapsto$$

$$-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}$$

$$0$$

$$\frac{\partial}{\partial \theta_2}$$

$$0$$

$$\mapsto$$

$$0$$

$$-q_2 \frac{\partial}{\partial p_2} + p_2 \frac{\partial}{\partial q_2}$$

$$T^2 \hookrightarrow T^2 \times \mathbb{R}^2 \cong_{T^2} (\mathbb{R}^2 \setminus \{0\})^2$$

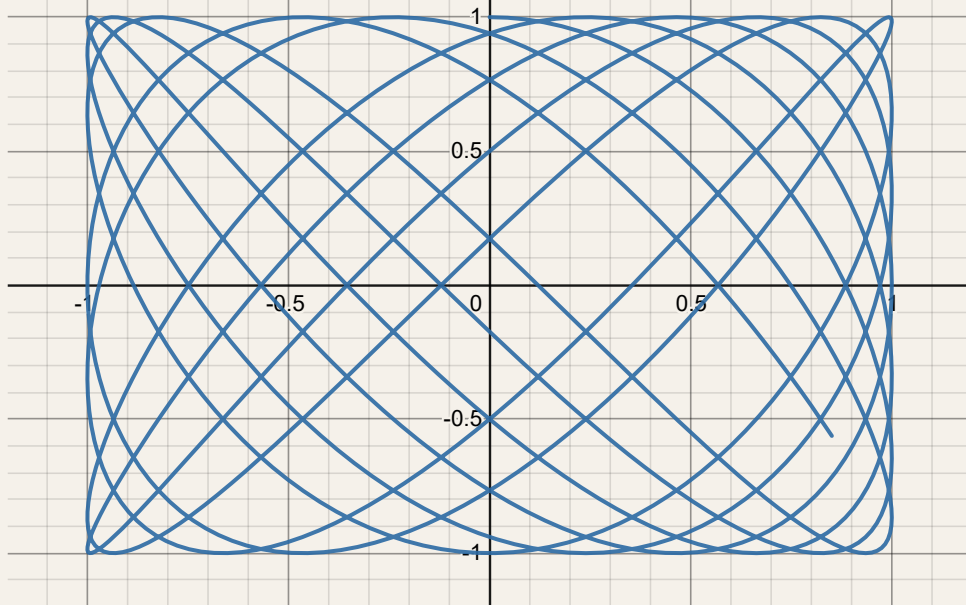
$$v_1 \mapsto \frac{\partial}{\partial \theta_1} \times 0 \mapsto X_1$$

$$v_2 \mapsto \frac{\partial}{\partial \theta_2} \times 0 \mapsto X_2$$

projecting onto (q_1, q_2)



$$\{(\sin 45t, \cos 65t) | t \in [0, 1]\}$$



Projecting $(0, q_1, 0, q_2)$ we have

$$(q_1, q_2) \xrightarrow{t} (\sin(\omega_1 t)q_1, \sin(\omega_2 t)q_2)$$

[1]

rational oscillator when $\omega_1, \omega_2 \in \mathbb{Z}$

A $S^1 \cong U(1)$ -action (or representation) on $\mathbb{R}^4 = \mathbb{C}^2$

irrational oscillator when $\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$

as a Lax pair

as a Lax pair

We define

$$L := \begin{bmatrix} p & q \\ q & -p \end{bmatrix}$$
$$M := \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

then

$$\begin{aligned} [M, L] &= \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} p & q \\ q & -p \end{bmatrix} - \begin{bmatrix} p & q \\ q & -p \end{bmatrix} \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\ &= \frac{\omega}{2} \begin{bmatrix} -q & p \\ p & q \end{bmatrix} - \frac{\omega}{2} \begin{bmatrix} q & -p \\ -p & -q \end{bmatrix} \\ &= \omega \begin{bmatrix} -q & p \\ p & q \end{bmatrix} \end{aligned}$$

Hence, the oscillator is equivalent to

$$\dot{L} = [M, L]$$

thus giving us conserved quantities

$$\mathrm{tr}(L^2) = \mathrm{tr}\left(\begin{bmatrix} p^2 + q^2 & ? \\ ? & 0 \end{bmatrix}\right) = p^2 + q^2$$

$$L := \begin{bmatrix} p_1 & q_1 \\ q_1 & -p_1 \end{bmatrix} + \begin{bmatrix} p_2 & q_2 \\ q_2 & -p_2 \end{bmatrix} z$$

$$M := \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$[M, A_1 + A_2 z] = [M, A_1] + [M, A_2] z$$

then

$$\dot{L} = [M, L]$$

should give the equation for isotropic harmonic oscillator.

Now

$$yI_2 - A_1 - A_2 z = \begin{bmatrix} y - p_1 - zp_2 & -q_1 - zq_2 \\ -q_1 - zq_2 & y + p_1 + zp_2 \end{bmatrix}$$

$$\begin{aligned} \det(yI_2 - A_1 - A_2 z) &= y^2 - (p_1 + zp_2)^2 - (q_1 + zq_2)^2 \\ &= y^2 - (p_1^2 + q_1^2) - (p_2^2 + q_2^2)z^2 - 2z(p_1p_2 + q_1q_2) \end{aligned}$$

which is a conic section over which

$$(y, z) \mapsto \ker(yI_2 - A(z))$$

defines a line bundle.

[2]

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1. <https://www.desmos.com/calculator/irnyqunfsm> ↩
 2. <https://www.desmos.com/calculator/2n2puejwvk> ↩

